

GeotagManager

QGIS Plugin — User Guide

Version 1.2.6

1. Installation

Download the latest **.zip** from the Releases page and install via **Plugins → Manage and Install Plugins → Install from ZIP**.

After installation, enable the plugin and open it via **Plugins → GeotagManager**.

ExifTool is required for GPX geotagging and EXIF writing. Install it from the plugin toolbar:  ExifTool → Download ExifTool.

2. Standard Workflow

Step	Action	Description
1	 Load Folder	Select the folder containing your photos
2	 Load GPX	Load one or more GPX tracks from the same session
3	Clock offset	Correct camera clock drift (seconds)
4	 Geotag from GPX	Match photos to trackpoints by timestamp
5	Review results	Green = matched, Orange = not matched
6	 Export GeoPackage	Save a point layer with all photo metadata

3. Interface Overview

3.1 Toolbar

The main toolbar provides access to all primary actions:

Button	Function
 Load Folder	Load photos from a directory (recursive)
 Load GPX	Load GPX track files
 Export GeoPackage	Export matched photos to a GPKG point layer
 Write EXIF on export	Also write GPS to original photo files on export
 ExifTool	Download/manage the bundled ExifTool binary

Button	Function
■ Log	Open the QGIS log panel mirror
■ Batch Scan → GPKG	Scan a directory tree and export without loading
■ Photo	Toggle full-screen photo preview mode

3.2 Left Panel

Sync from layer selection — connect GeotagManager to any QGIS point layer. When a point is selected on the map, the matching photo is loaded automatically.

Photos — list of loaded photos with their match status. Supports multi-selection (Shift+click, Ctrl+click). Click ■ **Open folder** to open the folder in the file manager.

GPX Tracks — list of loaded GPX files. Shows the total trackpoint count. The ■ **Geotag** button starts the matching process.

3.3 Right Panel — Photo Info

Shows EXIF metadata for the selected photo: filename, source, author, date/time, altitude, direction, camera model, and FL/HFOV. Read from the photo file on load; updated from the connected layer if available.

3.4 Right Panel — Edit Coordinates

Control	Description
■ Move Point	Click on the QGIS map to assign GPS coordinates. Supports sequential multi-photo mode.
■ Set Direction	Click the subject on the map to compute and write the camera azimuth (requires GPS).
Lat / Lon / Alt	Manual coordinate entry via spinboxes.
✓ Apply	Assign spinbox coordinates to all selected photos and update the connected layer.
— Write EXIF	Write GPS tags to original photo files (batch, with confirmation).

3.5 Controls Bar

Located at the bottom of the dialog:

Clock offset (sec) — Camera clock correction. If the camera was 5 minutes ahead of GPS time, enter -300.

Max gap (sec) — Skip photos if the surrounding GPX trackpoints are more than N seconds apart. Set to 0 for no limit.

■ **Assign authors** — Opens the Author Assignment dialog (requires a connected layer).

■ **Apply layer symbology** — Automatically applies green/red/wedge symbology when a layer is loaded or connected.

4. Features

4.1 Geotag from GPX

Each photo is matched to the nearest GPX trackpoint by timestamp. The GPS coordinates are interpolated between the two surrounding trackpoints.

After geotagging a summary dialog shows:

- ✓ Matched to GPX: N photos (green in list)
- ■ Outside GPX time range: N photos (orange in list)

4.2 Direction Tool

The ■ **Set Direction** tool computes the true azimuth from the camera position to a clicked target point using the geodetic formula. The result (0–360°, True North) is written to both the **GPSTimeDirection** EXIF tag and the **direction** field in the connected layer.

4.3 Multi-Selection Move Point

Select multiple photos with Shift+click, then press ■ **Move Point**. The tool stays active and processes photos sequentially — each click on the map assigns coordinates to the next photo in the selection. The status bar shows the current photo and remaining count.

4.4 Assign Authors

Opens a dialog that reads all features from the connected layer and groups them by **Date (day)** or **Camera**. Enter an author name for each group. Features that already have an author are not modified. Updates are written directly to the layer via **changeAttributeValues()**.

4.5 Export GeoPackage

The default filename is derived from photo EXIF dates:

Photos	Default filename	Layer name
Same day	20240226_layer.gpkg	20240226_photo_points
Multiple days	Start20240226_End20240228_layer.gpkg	Start20240226_End20240228_photo_points
No dates	geotagged_photos.gpkg	photo_points

After export, the layer is loaded automatically into the QGIS project and the map zooms to its extent.

5. GeoPackage Schema

Field	Type	Description
filename	String	Photo filename
filepath	String	Full path to the original file
author	String	Author (set via Assign Authors dialog)
datetime_photo	String	EXIF date/time (YYYY:MM:DD HH:MM:SS)
photo_date	Date	Date only — for temporal filtering in QGIS
latitude	Double	WGS84 latitude
longitude	Double	WGS84 longitude

Field	Type	Description
altitude	Double	Altitude in metres (ellipsoidal)
direction	Double	Camera azimuth 0–360° True North
pdop	Double	GPS dilution of precision
focal_length	Double	Focal length in mm
focal_length_35mm	Double	35mm equivalent focal length
hfov	Double	Horizontal field of view (degrees)
camera	String	Camera make + model
satellites	Integer	Number of GPS satellites
source	String	gpx exif manual pending skipped
notes	String	Free text notes

6. ExifTool Setup

GeotagManager bundles ExifTool for EXIF writing and GPX geotagging. The binary is installed in the plugin's **vendor/** folder — no system installation required.

To install: click **ExifTool** in the toolbar, then **Download ExifTool**. The plugin downloads the appropriate binary for your platform and extracts it automatically.

Status bar label	Meaning
■ ExifTool 13.54 (bundled)	Full functionality: GPX geotagging + EXIF write
■ piexif (JPEG only)	Limited: no GPX geotagging, JPEG-only EXIF write
■ Pure Python (built-in)	Read-only: no EXIF writing possible

7. Troubleshooting

Problem	Solution
Photos not matched after Geotag	Check clock offset. Compare photo timestamp with GPX time. Increase Max gap if GPS had d
ExifTool label is orange	Open ■ ExifTool and click Download. Ensure vendor/exiftool.exe and vendor/exiftool_files/ ex
Layer not loaded after Export	Open the .gpkg manually via Layer → Add Vector Layer. Check the QGIS Log panel for errors
Open Folder button greyed out	Load a photo folder first using ■ Load Folder, or connect a layer that triggers auto-loading.
Move Point does not update layer	The layer must be connected and must have filepath, latitude, longitude fields.
Direction not written to EXIF	ExifTool must be installed. Check the EXIF engine label in the status bar.